Customer Shopping Behavior — Executive Summary

Analysis of 3,900 transactions to reveal spending patterns, product preferences, subscription behavior, and customer segments to guide strategic decisions.



Made with GAMMA

Dataset Snapshot

Rows & Columns

3,900 rows · 18 columns

Key Features

Demographics, purchase details, behavior, subscription, shipping, ratings

Missing Data

37 nulls in Review Rating — imputed with category median



Made with GAMMA

Data Preparation & Engineering

Loaded with pandas; inspected using df.head(), df.info(), df.describe(). Standardized column names to snake_case. Imputed missing ratings by product category. Dropped redundant promo_code_used after consistency check.

Engineered features: age_group and purchase_frequency_days. Exported cleaned DataFrame to MySQL.



Made with GAMMA

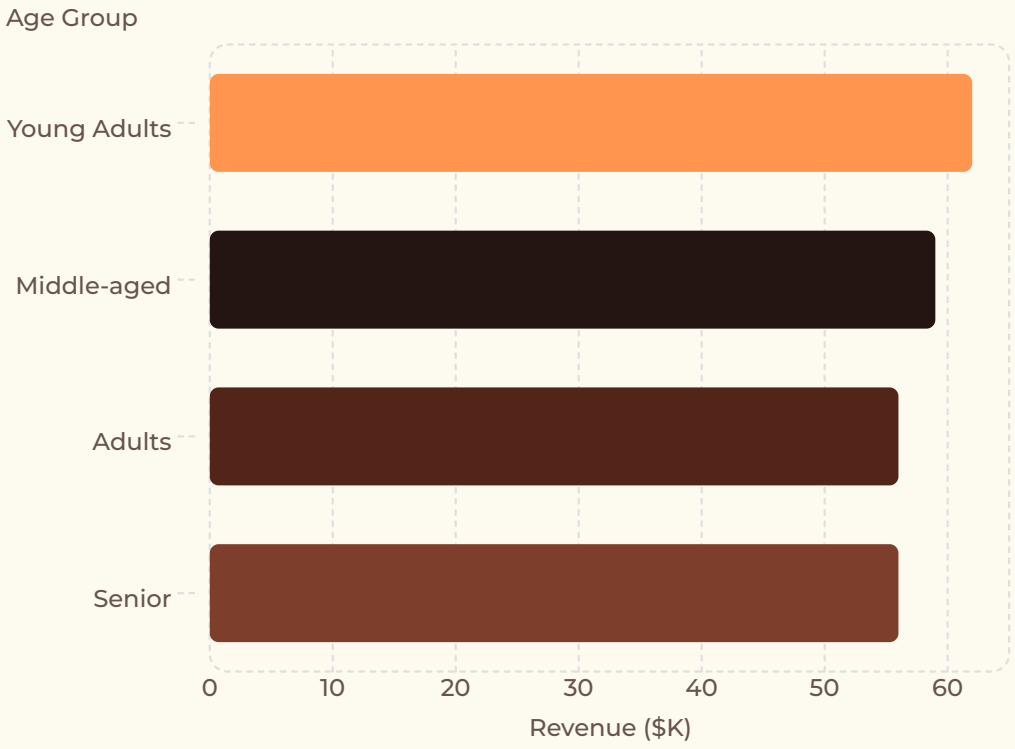


Top-Level Metrics

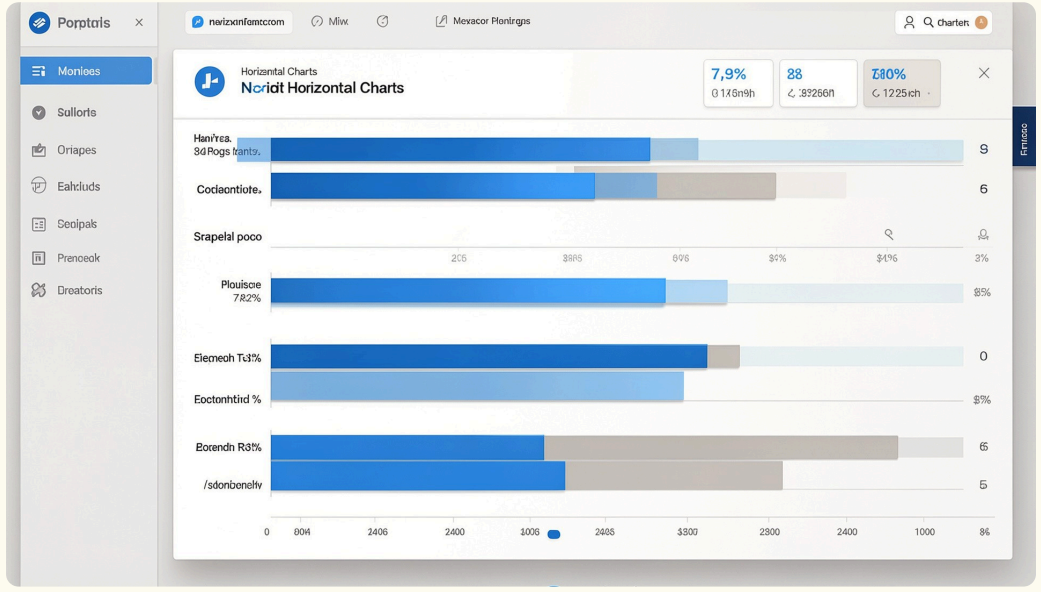
Customers 3,900 total	**Avg Purchase** $59.76
Avg Rating 3.75 (after imputation)	**Subscription Rate** 27% subscribed · 73% not subscribed

Made with GAMMA

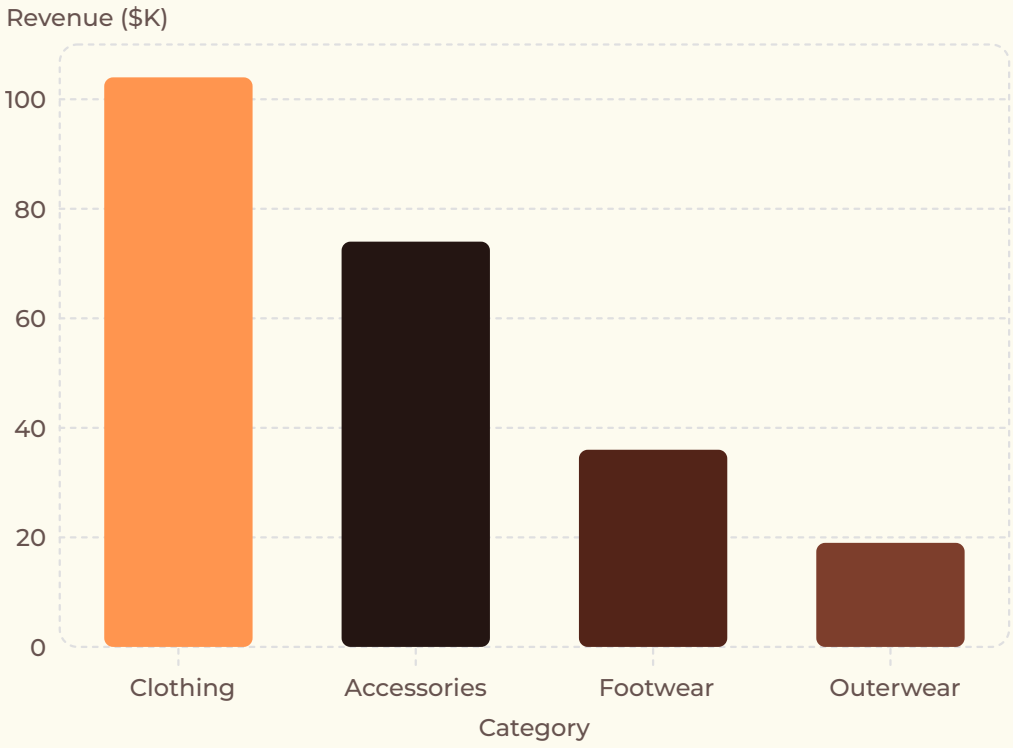
Revenue & Demographics



Revenue is concentrated in Young Adults and Middle-aged groups. Segment marketing to these cohorts for greater ROI.



Category Performance



Top by revenue: Clothing leads. By unit sales: Clothing (1,737), Accessories (1,240), Footwear (599), Outerwear (324). Use promotions to protect margin while driving volume.



Made with GAMMA

Product & Rating Highlights



Gloves — 3.86



Sandals — 3.84



Boots — 3.82



Hat — 3.80



Handbag — 3.78



Skirt — 3.78

Made with GAMMA

Behavioral Patterns & Segments

1 Customer Segments

Loyal: 3,116 · Returning: 701 · New: 83

2 Subscribers vs Non

Subscribers: 1,053 (avg $59.49, total $62,645). Non-subscribers: 2,847 (avg $59.87, total $170,436).

3 Discount Dependence

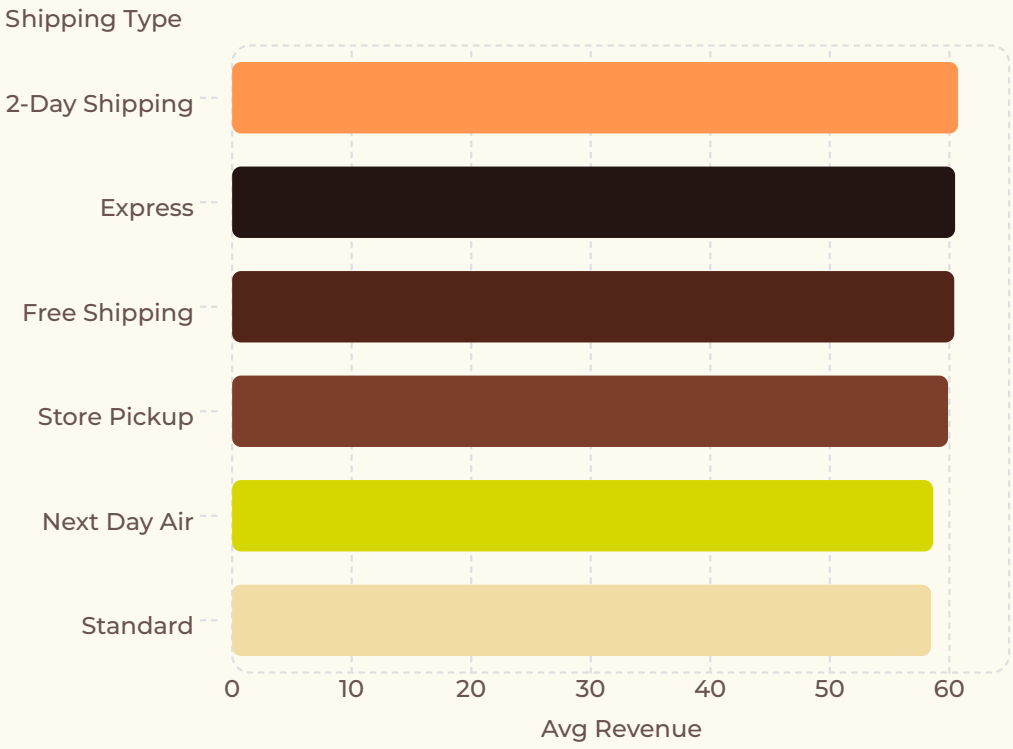
Top discount-rate items: Hat 50%, Sneakers 49.66%, Coat 49.07%, Sweater 48.17%, Pants 47.37%.



839 customers used discounts yet spent above average — indicates high-value discount users worth targeted loyalty offers.

Made with GAMMA

Shipping & Payment Insights



Minimal variation across shipping modes. Payment method top frequency: PayPal (most common). Consider optimizing shipping-cost strategies and promoted shipping options tied to conversion.



Made with GAMMA

Recommendations & Next Steps

Boost Subscriptions

Promote exclusive benefits and targeted messaging (focus female cohort) to raise conversion.

Loyalty Program

Reward returning customers to move them to Loyal segment and increase lifetime value.

Discount Policy Review

Balance acquisition vs margin; identify high-value discount users for tailored offers.

Promote Top Products

Market top-rated and top-selling items (Clothing, Gloves, Sandals) across age cohorts.

Targeted Marketing

Prioritize Young Adults & Middle-aged segments and preferred shipping modes for campaigns.

Dashboard available in Power BI for interactive exploration. Next: A/B test subscription incentives and loyalty mechanics; monitor lift in revenue and retention.



Made with GAMMA